

Lab #6 [OEL]

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#include <DHT.h>

#define DHTPIN D2
#define DHTTYPE DHT11
#define LEDPIN D6

DHT dht(DHTPIN, DHTTYPE);

float tempThreshold = 30.0;
float humThreshold = 40.0;

void setup() {
  Serial.begin(115200);
  pinMode(LEDPIN, OUTPUT);

  // External LED starts OFF (Standard Logic)
  digitalWrite(LEDPIN, LOW);
  dht.begin();
}

void loop() {
  float temp = dht.readTemperature();
  float hum = dht.readHumidity();

  // Safety: If sensor fails, keep LED OFF to avoid false alerts
  if (isnan(temp) || isnan(hum)) {
    Serial.println("Sensor Error!");
    digitalWrite(LEDPIN, LOW);
    delay(2000);
    return;
  }

  Serial.print("Temp: "); Serial.print(temp);
  Serial.print("C | Hum: "); Serial.print(hum);
  Serial.println("%");

  // Threshold Logic (Active-High for External LED)
  if (temp > tempThreshold || hum > humThreshold) {
    digitalWrite(LEDPIN, HIGH); // External LED ON
    Serial.println("Threshold Exceeded: LED ON");
  } else {
    digitalWrite(LEDPIN, LOW); // External LED OFF
    Serial.println("Normal: LED OFF");
  }

  Serial.println("-----");
  delay(2000);
}
```

```
Temp: 23.80C | Hum: 41.00%
Threshold Exceeded: LED ON
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Temp: 23.80C | Hum: 41.00%
Threshold Exceeded: LED ON
-----
Temp: 23.80C | Hum: 41.00%
Threshold Exceeded: LED ON
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